Deep Convolutional Neural Network Analysis of Biomechanical Gait Improvements Following Ankle-Foot Orthosis use in Stroke Patients

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Introduction

Advanced computational approaches, such as deep convolutional neural networks (DCNN), provide new opportunities for objectively classifying and interpreting complex biomechanical gait improvements following Ankle-Foot Orthosis (AFO) use in stroke rehabilitation. This study aimed to evaluate the efficacy of DCNN models in distinguishing affected versus control gait patterns and identifying subtle biomechanical improvements after AFO use, utilizing Gradient-weighted Class Activation Mapping (Grad-CAM) for interpretability.

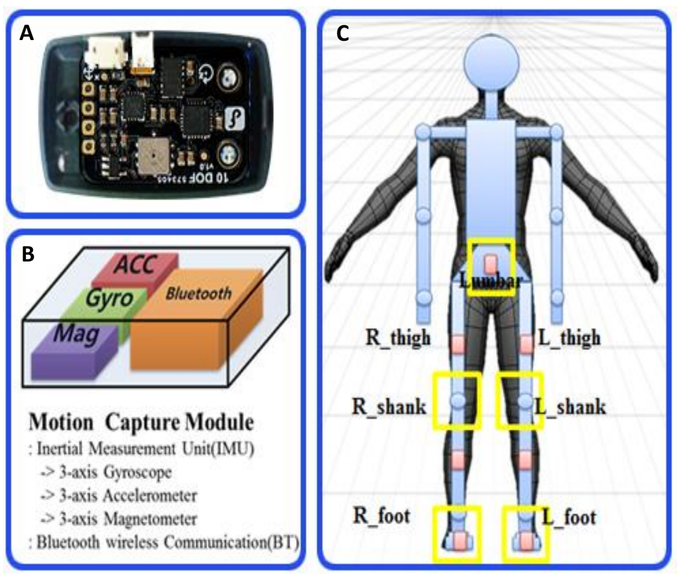

Figure 1. IMU-based motion capture and sensor placement. (A) Example of the wireless inertial measurement unit (IMU) used. (B) Module architecture integrating a tri-axial accelerometer, gyroscope, and magnetometer with Bluetooth communication.(C) Body placement of seven sensors (yellow boxes)

Methods

Gait data from 48 stroke patients (56 datasets) were collected pre- and post-AFO using inertial measurement units. Additionally, an extensive control dataset comprising 5,826 gait samples from 828 healthy individuals, previously validated was included to train and validate the DCNN model. Raw IMU sensor data underwent rigorous preprocessing including normalization, alignment, and segmentation into fixed-length sequences. Multi-plane hip-knee cyclogram data were transformed into numerical arrays representing sagittal, coronal, and transverse joint angles. Clinical covariates including age, sex, height, and weight underwent Z-score normalization for standardization.

The DCNN model was developed and validated for two primary tasks: (1) distinguishing affected stroke gait from normal gait patterns, and (2) detecting subtle biomechanical gait improvements post-AFO. Grad-CAM visualizations identified critical gait phases significantly enhanced by AFO use.

Results

The DCNN model achieved exceptional accuracy (99.9%), precision (100%), recall (100%), and F1-score (100%).

Table 1. Classification reports for subject-wise results

	Precision	Recall	F1-score	Support
Control	1.00	1.00	1.00	829
Affected	0.96	1.00	0.98	27
Average	0.98	1.00	0.99	856

Grad-CAM visualizations highlighted key biomechanical improvements, particularly increased hip and knee flexion during initial swing, improved knee extension and hip stability at terminal swing to initial contact transition, and enhanced joint stability during mid-stance phases. . Notably, only two of 28 patients exhibited gait patterns approaching normal following AFO use, indicating a need for individualized rehabilitation strategies beyond orthotic support.

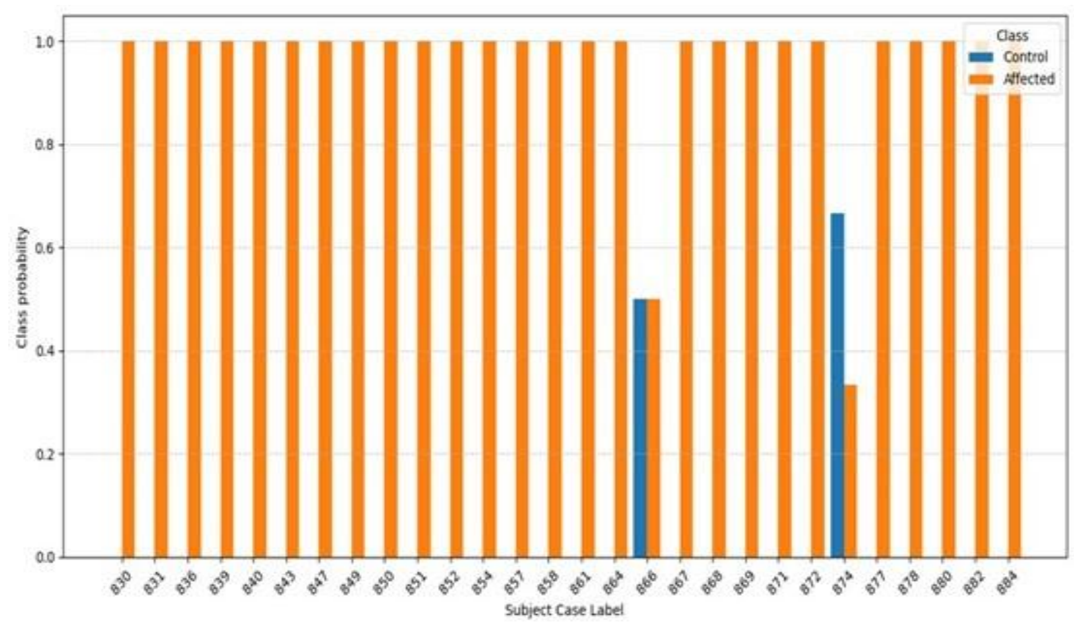

Figure 2. Class probabilities of gait samples after AFO use. Most samples remain classified as "affected gait" (orange bars); two samples (#866, #874) show notable shifts toward "control gait" (blue bars)

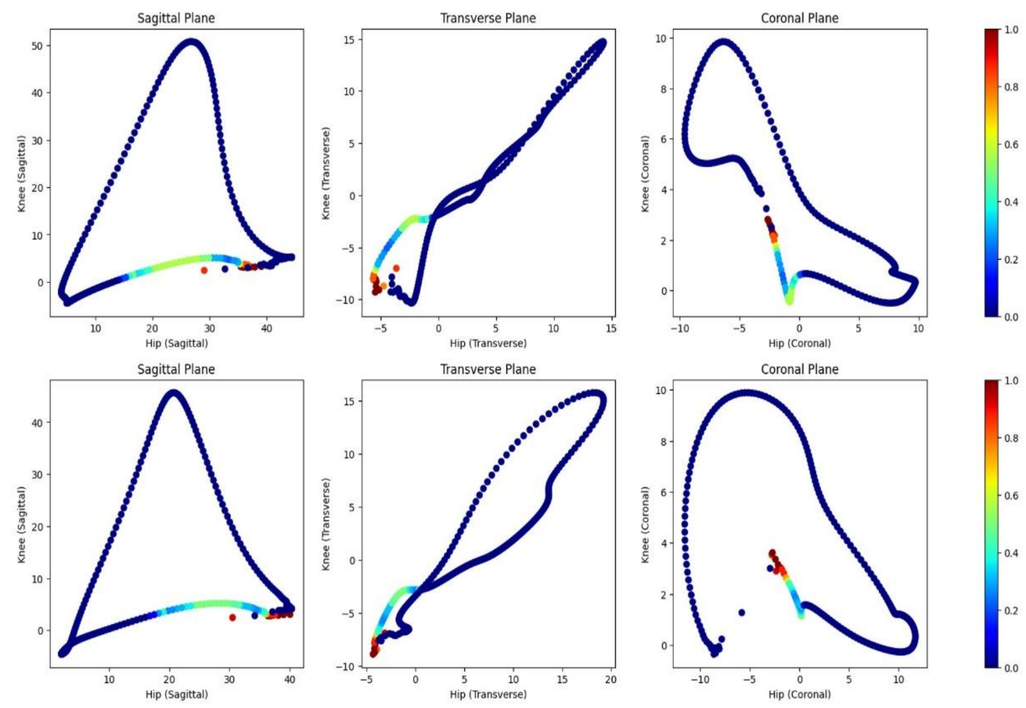

Figure 3. Grad-CAM visualization of two gait cycles from sample #874 using hip-knee cyclograms. Intervals highlighted in red represent critical regions identified by the DCNN model when classifying gait improvements.

Conclusion

The DCNN analysis successfully identified and visualized clinically relevant biomechanical gait improvements, underscoring its utility for precision medicine and individualized stroke rehabilitation strategies.

Acknowledgement

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